

# claude-windows / 188dd06f-9c99-487c-aa7d-b0bee83c629c

## Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\188dd06f-9c99-487c-aa7d-b0bee83c629c\subagents\workflows\wf\_416cc0e5-a35\agent-a95c2b6dfbe328ed5.jsonl`
- SHA-256: `a8835090b7afc25c99d168103e0dd0663378600a29058247c426d8ed748c8910`
- Source modified: `2026-06-22T09:50:02+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf\_416cc0e5-a35`
- session\_id: `188dd06f-9c99-487c-aa7d-b0bee83c629c`

## Transcript

### 1. user (2026-06-22T09:47:21.573Z)

Review the NIGHTLY REEL-COUNT engine-regression change under C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/nightly-e2e (a NEW, additive PR). It runs the configured WASB pipeline on a GCP GPU VM nightly, asserts reel-count (==) + quality metrics + cost, emails a report, self-deletes. Files: scripts/nightly\_reelcount.py, scripts/nightly\_email.py, deploy/gcp/nightly\_regression/run\_on\_vm.sh, deploy/gcp/nightly\_regression/launch.sh, deploy/gcp/nightly\_regression/register\_scheduler.sh, deploy/gcp/nightly\_regression/Dockerfile.launcher, eval\_baselines/reelcount\_manifest.json. Read them under that absolute path. Cite file:line. Report only REAL defects (correctness/safety/security that would bite in production or a regression that escapes), not style. Return raw structured findings.

ADVERSARIALLY VERIFY this finding ? confirm with concrete code evidence under C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/nightly-e2e or REFUTE it (default real=false if the evidence isn't clearly there). Finding:

```
{
  "title": "No error handling around SMTP login ? password may leak in exception trace",
  "file": "scripts/nightly_email.py",
  "line": "86-89",
  "severity": "medium",
  "detail": "The `send_report()` function does not wrap `s.login(user, password)` in a try-except block. If SMTP login fails (e.g., wrong password, server error), Python's default exception handler will raise an smtplib.SMTPAuthenticationError, which may include the username and password in the exception message/traceback when logged or printed. The surrounding code catches the parent exception elsewhere in the orchestration (nightly_reelcount.py line 539-540), but the password string object exists in the traceback.",
  "fix": "Wrap the SMTP operations in a dedicated try-except that catches `smtplib.SMTPException`, logs a sanitized message (no credentials), and re-raises or returns False gracefully. Example: `except smtplib.SMTPAuthenticationError: log.error('SMTP login failed'); return False`"
}
```

### 2. assistant (2026-06-22T09:47:23.878Z)

I'll analyze the NIGHTLY REEL-COUNT engine-regression change, focusing on adversarially verifying the finding about SMTP login error handling and password leakage. Let me start by reading the relevant files.

### 3. user (2026-06-22T09:47:24.873Z)

```
1      """Email the nightly regression report ? stdlib SMTP, creds from env or GCP Secret
Manager.
2
3      Used by ``scripts/nightly_reelcount.py --email`` (and reusable by any nightly).
Deliberately
```

```

4      dependency-light (stdlib ``smtplib`` + ``email.message``) and **graceful**: if no SMTP
password can
5      be resolved it logs + returns ``False`` (the report is still written/printed ? a broken
mailer must
6      never mask a regression).
7
8      Credential resolution (first hit wins), so it works on a GCP VM, locally, or in tests:
9      ? Recipient: ``--email-to`` arg ? ``NIGHTLY_EMAIL_TO`` ? ``avi.dullu@gmail.com``
10     ? SMTP host: ``NIGHTLY_SMTP_HOST`` (default ``smtp.gmail.com``) . port
``NIGHTLY_SMTP_PORT`` (587, STARTTLS)
11     ? SMTP user: ``NIGHTLY_SMTP_USER`` (default = recipient)
12     ? SMTP pass: ``NIGHTLY_SMTP_PASS`` ? Secret Manager ``NIGHTLY_SMTP_SECRET``
13     (``projects/<p>/secrets/<name>/versions/latest``; via the google-cloud-
secret-manager
14     lib if installed, else the ``gcloud`` CLI). A Gmail **app password** is
the simplest.
15
16     Set the secret once (owner): echo -n '<app-password>' | gcloud secrets create nightly-
smtp-pass --data-file=-
17     and on the VM: export NIGHTLY_SMTP_SECRET=projects/<proj>/secrets/nightly-smtp-
pass/versions/latest
18     export NIGHTLY_SMTP_USER=<your-gmail>
NIGHTLY_EMAIL_TO=avi.dullu@gmail.com
19     ""
20     from __future__ import annotations
21
22     import logging
23     import os
24     from email.message import EmailMessage
25     from typing import Optional
26
27     log = logging.getLogger("nightly_email")
28
29     DEFAULT_TO = "avi.dullu@gmail.com"
30     DEFAULT_HOST = "smtp.gmail.com"
31     DEFAULT_PORT = 587
32
33
34     def build_message(subject: str, body_md: str, from_addr: str, to_addr: str) ->
EmailMessage:
35     """A text/plain email carrying the markdown report (renders fine in any client).
Pure ? testable."""
36     msg = EmailMessage()
37     msg["Subject"] = subject
38     msg["From"] = from_addr
39     msg["To"] = to_addr
40     msg.set_content(body_md)
41     return msg
42
43
44     def resolve_secret(env: Optional[dict] = None) -> Optional[str]:
45     """Resolve the SMTP password from env or GCP Secret Manager. Returns None if
unresolvable."""
46     env = env if env is not None else os.environ
47     direct = env.get("NIGHTLY_SMTP_PASS")
48     if direct:
49         return direct
50     secret_ref = env.get("NIGHTLY_SMTP_SECRET")
51     if not secret_ref:
52         return None
53     # Prefer the client lib; fall back to the gcloud CLI (no extra dependency required).
54     try:
55         from google.cloud import secretmanager # type: ignore
56         client = secretmanager.SecretManagerServiceClient()
57         return

```

```

client.access_secret_version(name=secret_ref).payload.data.decode("utf-8")
58     except Exception: # noqa: BLE001 ? lib absent or auth issue ? try the CLI
59         pass
60     try:
61         import subprocess
62         name = secret_ref.split("/secrets/", 1)[1].split("/", 1)[0] if "/secrets/" in
secret_ref else secret_ref
63         out = subprocess.run(["gcloud", "secrets", "versions", "access", "latest", "--
secret", name],
64                               capture_output=True, text=True, timeout=30)
65         return out.stdout.strip() if out.returncode == 0 and out.stdout.strip() else
None
66     except Exception as e: # noqa: BLE001
67         log.warning("secret-manager resolution failed: %s", e)
68         return None
69
70
71     def send_report(subject: str, body_md: str, to: Optional[str] = None,
72                   env: Optional[dict] = None) -> bool:
73         """Send the report. Returns True if sent, False if skipped (no creds) ? never raises
on missing creds."""
74         env = env if env is not None else os.environ
75         to_addr = to or env.get("NIGHTLY_EMAIL_TO") or DEFAULT_TO
76         host = env.get("NIGHTLY_SMTP_HOST", DEFAULT_HOST)
77         port = int(env.get("NIGHTLY_SMTP_PORT", DEFAULT_PORT))
78         user = env.get("NIGHTLY_SMTP_USER") or to_addr
79         password = resolve_secret(env)
80         if not password:
81             log.warning("no SMTP password (set NIGHTLY_SMTP_PASS or NIGHTLY_SMTP_SECRET) ?
email skipped.")
82             return False
83
84         import smtplib
85         msg = build_message(subject, body_md, from_addr=user, to_addr=to_addr)
86         with smtplib.SMTP(host, port, timeout=60) as s:
87             s.starttls()
88             s.login(user, password)
89             s.send_message(msg)
90         log.info("emailed report to %s", to_addr)
91         return True
92

```

#### 4. user (2026-06-22T09:47:26.185Z)

```

1     #!/usr/bin/env bash
2     # Startup-script for the NIGHTLY engine-regression VM. Runs UNATTENDED on boot, then
DELETES the VM
3     # (cost rule: a running GPU VM is the only real cost). Survives SSH drops (it's a
startup-script, not
4     # an interactive session). Reads its config from instance metadata (set by launch.sh).
5     #
6     # Flow: Ops Agent ? pull repo @ HEAD from GCS ? build venv + WASB env ? stage
weights ?
7     #         run scripts/nightly_reelcount.py on the gs:// golden clips (emails the report)
?
8     #         upload report to GCS ? SELF-DELETE (even on failure, via the EXIT trap).
9     #
10    # Metadata keys (launch.sh sets these): repo-tgz-uri, git-sha, gcs-bucket, report-
prefix,
11    # weights-uri, email (true/false). NOT verified in CI ? dry-run via launch.sh on a
real VM first.
12    set -uo pipefail
13    exec > >(tee -a /var/log/nightly-regression.log) 2>&1
14    echo "==== nightly-regression startup $(date -u +%FT%TZ) ====="
15    BOOT_EPOCH=$(date +%s)

```

```

16
17     md() { curl -s -H "Metadata-Flavor: Google"
"http://metadata.google.internal/computeMetadata/v1/$1"; }
18     NAME=$(md instance/name)
19     ZONE=$(md instance/zone | awk -F/ '{print $NF}')
20     REPO_TGZ=$(md instance/attributes/repo-tgz-uri || true)
21     GIT_SHA=$(md instance/attributes/git-sha || echo unknown)
22     BUCKET=$(md instance/attributes/gcs-bucket || true)
23     REPORT_PREFIX=$(md instance/attributes/report-prefix || echo nightly/reports)
24     WEIGHTS_URI=$(md instance/attributes/weights-uri || true)
25     DO_EMAIL=$(md instance/attributes/email || echo true)
26     SMTP_SECRET=$(md instance/attributes/smtp-secret || true)
27     SMTP_USER=$(md instance/attributes/smtp-user || true)
28     EMAIL_TO=$(md instance/attributes/email-to || echo avi.dullu@gmail.com)
29     DATE=$(date -u +%F)
30
31     # ALWAYS delete the VM on exit (success OR failure) ? never leave a billing GPU VM
behind.
32     cleanup() {
33         echo "==== self-delete $NAME ($ZONE) ====="
34         gcloud compute instances delete "$NAME" --zone="$ZONE" --quiet || \
35             echo "WARN: self-delete failed ? DELETE MANUALLY: gcloud compute instances delete
$NAME --zone=$ZONE -q"
36     }
37     trap cleanup EXIT
38
39     fail_email() { # best-effort failure alert (the report email won't have fired on an
early crash)
40         echo "RUN FAILED ? attempting a failure alert"
41         if [ "$DO_EMAIL" = "true" ] && [ -d ~/rally ]; then
42             cd ~/rally 2>/dev/null && . .venv/bin/activate 2>/dev/null && \
43                 NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
44                 python - <<PY || true
45         from scripts.nightly_email import send_report
46         send_report("[KhelSutra nightly] ERROR ? engine regression VM run failed ($DATE)",
47             "The nightly engine-regression run failed on the VM before producing a
report.\n"
48             "See gs://$BUCKET/$REPORT_PREFIX/$DATE/ and the VM serial
log.\nGIT_SHA=$GIT_SHA", to="$EMAIL_TO")
49         PY
50         fi
51     }
52     trap 'fail_email' ERR
53
54     # 1) Observability (Ops Agent) ? same as create_gpu_vm.sh bakes in.
55     curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh && \
56         sudo bash add-google-cloud-ops-agent-repo.sh --also-install || echo "WARN: ops-agent
install failed (non-fatal)"
57
58     # 2) Pull the repo @ HEAD (staged to GCS by launch.sh ? no GitHub auth needed; uses the
worker SA).
59     mkdir -p ~/rally && cd ~/rally
60     gcloud storage cp "$REPO_TGZ" /tmp/rally.tgz
61     tar -xzf /tmp/rally.tgz -C ~/rally
62     echo "$GIT_SHA" > ~/rally/GIT_SHA
63
64     # 3) Build the env + native WASB (provider-agnostic DO scripts, reused on GCP per
deploy/gcp/README $4).
65     sudo bash deploy/digitalocean/mount_scratch.sh 2>/dev/null || true
66     export TMPDIR=/scratch 2>/dev/null || true
67     ./deploy/digitalocean/bootstrap.sh --gpu
68     . .venv/bin/activate
69     pip install -e '[storage-gcs]'
70     bash deploy/digitalocean/gpu_setup.sh || echo "WARN: gpu_setup (WASB env) returned

```

```

nonzero ? continuing"
71
72 # 4) Stage the WASB checkpoint (gpu_setup does NOT fetch weights ? README §4).
73 if [ -n "$WEIGHTS_URI" ]; then
74     mkdir -p ~/models/WASB-SBDT/pretrained_weights
75     gcloud storage cp "$WEIGHTS_URI" ~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar
76     export WASB_WEIGHTS=~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
77 fi
78
79 # 5) Run the regression on the gs:// golden clips in the committed manifest. Cost = VM
uptime so far.
80 UPTIME=$(( $(date +%s) - BOOT_EPOCH ))
81 EMAIL_FLAG=""; [ "$DO_EMAIL" = "true" ] && EMAIL_FLAG="--email --email-to $EMAIL_TO"
82 set +e
83 NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
84 python scripts/nightly_reelcount.py --vm-uptime-seconds "$UPTIME" $EMAIL_FLAG
85 RUN_RC=$?
86 set -e
87 echo "runner exit code: $RUN_RC (0 pass / 1 regression / 4 stale)"
88
89 # 6) Upload the dated report to GCS for the record (alongside the email).
90 if [ -n "$BUCKET" ] && [ -d output/nightly_reelcount ]; then
91     gcloud storage cp -r "output/nightly_reelcount/$DATE.*"
"gs://$BUCKET/$REPORT_PREFIX/$DATE/" || \
92     gcloud storage cp -r output/nightly_reelcount "gs://$BUCKET/$REPORT_PREFIX/$DATE/"
|| true
93 fi
94
95 echo "==== nightly-regression done rc=$RUN_RC $(date -u +%FT%TZ) ====="
96 trap - ERR # disable the failure-email trap; the EXIT (self-delete) trap stays
97 # the EXIT trap deletes the VM
98

```

## 5. user (2026-06-22T09:47:26.283Z)

```

1      ""Nightly END-TO-END engine regression ? reel-count + quality + cost (the GCP-VM
nightly).
2
3      Runs the configured + checkpointed pipeline (the real WASB GPU detector, as a
deployment serves
4      it) end-to-end on 2-3 short owned golden videos and asserts the number of reels does
not change
5      vs a frozen baseline. Unlike ``scripts/nightly_regression.py`` (PR #202 ? a CPU re-score
of CACHED
6      trajectories, the windowing/scorer only), this one runs the full pipeline on the real
model, so it
7      needs a GPU ? hence it runs on an ephemeral GCP GPU VM
(``deploy/gcp/nightly_regression/``), emails the
8      owner the report, and self-deletes. It is default-additive: a tripwire, not the
promotion gate.
9
10     What it asserts / reports, per the owner's ask ("quality metrics + regression report +
cost"):
11     ? REEL COUNT per video ? ``len(play_segments)`` after segmentation; the detected
highlight clips.
12     Pure-CV / WASB detection is deterministic for a fixed (video, config, weights), so
this is asserted
13     EXACT (with a re-run-once guard for the one known flake: a transient DB-insert-
None, see §guard).
14     ? QUALITY METRICS per video (when golden labels are provided) ? the R2 tolerance-match
block from
15     ``backend.eval.rally_seg_eval`` (tol_f1 / f1@0.5 / f1@0.25), compared with a small
tolerance.

```

```

16      ? COST of the run ? VM uptime (or processing wall) × machine $/hr via
  ``backend.billing`` ? USD + INR.
17
18      Extensible pattern (add a video = one manifest row + re-freeze):
19      ``eval_baselines/reelcount_manifest.json`` ? the videos (gs:// / local / gdrive://
  URIs) + segmenter
20      ``eval_baselines/reelcount_baseline.json`` ? the frozen expected counts (regenerated,
  never hand-edited)
21
22      Usage
23      -----
24      Freeze the baseline from the current configured+checkpointed engine (after an intended
  change, or first run):
25      python scripts/nightly_reelcount.py --update-baseline
26
27      Nightly check (exit 1 on regression; writes output/nightly_reelcount/<date>.md; emails
  if --email):
28      python scripts/nightly_reelcount.py --email
29
30      Exit codes: 0 pass · 1 regression (reel-count changed / a video errored / quality
  dropped) · 2 operational
31      (manifest/videos missing, nothing ran) · 3 no baseline · 4 stale (corpus/segmenter
  changed ? re-freeze).
32      """
33      from __future__ import annotations
34
35      import argparse
36      import datetime as _dt
37      import json
38      import os
39      import sys
40      from dataclasses import dataclass, field
41      from typing import Any, Dict, List, Optional, Tuple
42
43      REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
44      if REPO_ROOT not in sys.path:
45          sys.path.insert(0, REPO_ROOT)
46
47      # billing is pure (typing only) ? safe to import at module top.
48      from backend import billing # noqa: E402
49
50      DEFAULT_MANIFEST = os.path.join(REPO_ROOT, "eval_baselines", "reelcount_manifest.json")
51      DEFAULT_BASELINE = os.path.join(REPO_ROOT, "eval_baselines", "reelcount_baseline.json")
52      DEFAULT_REPORT_DIR = os.path.join(REPO_ROOT, "output", "nightly_reelcount")
53
54      #: Quality metrics we report + (when labels exist) gate ? higher is better, small
  tolerance.
55      QUALITY_HIGHER: Tuple[str, ...] = ("tol_f1", "f1@0.5", "f1@0.25")
56      DEFAULT_QUALITY_TOL = 0.02      # quality is float/labeled ? a looser tripwire than the
  exact reel-count
57      DEFAULT_FX_INR_PER_USD = 83.0
58      #: Fallback machine price if the manifest doesn't pin one (T4 on n1-standard-8, asia-
  south1, ~mid-2026).
59      DEFAULT_MACHINE_USD_PER_HR = 0.35
60
61
62      @dataclass(frozen=True)
63      class Tolerances:
64          quality_tol: float = DEFAULT_QUALITY_TOL
65
66          def to_dict(self) -> Dict[str, Any]:
67              return {"quality_tol": self.quality_tol}
68
69
70      # -----

```

```

71     # Pure cost math (wraps backend.billing ? no IO).
72     # ----- #
73     def cost_report(billed_seconds: float, machine_usd_per_hr: float,
74                     fx_inr_per_usd: float, gpu_seconds: float = 0.0) -> Dict[str, Any]:
75         """Run cost from billed wall-seconds × machine rate ? USD + INR (``backend.billing``
arithmetic).
76
77         ``billed_seconds`` is the VM uptime (boot?delete) when the orchestrator passes
``--vm-uptime-seconds``;
78         otherwise it's the in-process pipeline wall time (a lower bound ? the VM also pays
for boot/install)."""
79         rate_per_sec = float(machine_usd_per_hr) / 3600.0
80         usage = {billing.WALL_SECONDS: float(billed_seconds)}
81         rates = {billing.WALL_SECONDS: rate_per_sec}
82         usd, breakdown = billing.compute_cost_usd(usage, rates)
83         return {
84             "billed_seconds": round(float(billed_seconds), 1),
85             "gpu_seconds": round(float(gpu_seconds), 1),
86             "machine_usd_per_hr": float(machine_usd_per_hr),
87             "usd": usd,
88             "inr": billing.to_inr(usd, fx_inr_per_usd),
89             "breakdown": breakdown,
90         }
91
92
93     # ----- #
94     # Pure summary + comparison core (no IO ? unit-testable with plain dicts).
95     # ----- #
96     def summarize_results(run_results: List[Dict[str, Any]], segmenter: str,
97                           cost: Dict[str, Any]) -> Dict[str, Any]:
98         """Compact, comparable snapshot from per-video run results.
99
100         Each run result: ``{name, reel_count, ok, error, quality: {tol_f1,...}|None,
wall_seconds}``.
101         A video that errored keeps ``reel_count=None`` + ``ok=False`` (so the diff flags it,
never hides it)."""
102         per_video: Dict[str, Any] = {}
103         for r in run_results:
104             per_video[r["name"]] = {
105                 "reel_count": r.get("reel_count"),
106                 "ok": bool(r.get("ok", False)),
107                 "error": r.get("error"),
108                 "quality": r.get("quality"),
109             }
110         return {"segmenter": segmenter, "cost": cost, "per_video": per_video,
111               "n": len(per_video)}
112
113
114     @dataclass
115     class Finding:
116         video: str
117         metric: str # "reel_count" | "error" | a quality key | "segmenter" |
"corpus"
118         kind: str # "regression" | "improvement" | "stale"
119         baseline: Optional[float] = None
120         current: Optional[float] = None
121         detail: str = ""
122
123         def message(self) -> str:
124             if self.kind == "stale":
125                 return f"[STALE] {self.metric}: {self.detail}"
126             if self.metric == "error":
127                 return f"[REGRESSION] {self.video}: pipeline ERROR ? {self.detail}"
128             b = "?" if self.baseline is None else f"{self.baseline:g}"
129             c = "?" if self.current is None else f"{self.current:g}"

```

```

130         tag = "REGRESSION" if self.kind == "regression" else "improvement"
131         return f"[{tag}] {self.video}/{self.metric}: {b} ? {c}"
132
133
134     @dataclass
135     class NightlyResult:
136         findings: List[Finding] = field(default_factory=list)
137         added: List[str] = field(default_factory=list)
138         removed: List[str] = field(default_factory=list)
139
140         @property
141         def regressions(self) -> List[Finding]:
142             return [f for f in self.findings if f.kind == "regression"]
143
144         @property
145         def improvements(self) -> List[Finding]:
146             return [f for f in self.findings if f.kind == "improvement"]
147
148         @property
149         def stale(self) -> List[Finding]:
150             return [f for f in self.findings if f.kind == "stale"]
151
152         def overall_status(self) -> str:
153             if self.regressions:
154                 return "REGRESSION"
155             if self.stale:
156                 return "STALE"
157             return "PASS"
158
159         def exit_code(self) -> int:
160             if self.regressions:
161                 return 1
162             if self.stale:
163                 return 4
164             return 0
165
166
167     def compare(baseline: Dict[str, Any], current: Dict[str, Any], tols: Tolerances) ->
NightlyResult:
168         """Diff current run vs frozen baseline.
169
170         * Segmenter changed ? STALE (numbers not comparable; re-freeze).
171         * Corpus (video set) changed ? STALE + report added/removed; still diff the
INTERSECTION.
172         * Per shared video: reel_count EXACT (any change = regression); a pipeline ERROR =
regression;
173         quality metrics drop beyond ``quality_tol`` = regression (rise = improvement).
174         """
175         res = NightlyResult()
176         if baseline.get("segmenter") != current.get("segmenter"):
177             res.findings.append(Finding("", "segmenter", "stale",
178                                     detail=f"{baseline.get('segmenter')} ?
{current.get('segmenter')}"))
179             return res
180
181         b_pv, c_pv = baseline.get("per_video", {}), current.get("per_video", {})
182         res.added = sorted(set(c_pv) - set(b_pv))
183         res.removed = sorted(set(b_pv) - set(c_pv))
184         if res.added or res.removed:
185             res.findings.append(Finding("", "corpus", "stale",
186                                     detail=f"added={res.added} removed={res.removed}"))
187
188         for v in sorted(set(b_pv) & set(c_pv)):
189             b, c = b_pv[v], c_pv[v]
190             # 1) pipeline must still succeed

```



```

191         if not c.get("ok", False) or c.get("reel_count") is None:
192             res.findings.append(Finding(v, "error", "regression",
193                                     detail=str(c.get("error") or "no reel_count
produced"))))
194             continue
195         # 2) reel count ? EXACT
196         bc, cc = b.get("reel_count"), c.get("reel_count")
197         if bc is not None and cc is not None and cc != bc:
198             res.findings.append(Finding(v, "reel_count", "regression", float(bc),
float(cc)))
199         # 3) quality metrics ? tolerance (only if both sides have them)
200         bq, cq = b.get("quality") or {}, c.get("quality") or {}
201         for m in QUALITY_HIGHER:
202             bv, cv = bq.get(m), cq.get(m)
203             if bv is None or cv is None:
204                 continue
205             if cv < bv - tols.quality_tol:
206                 res.findings.append(Finding(v, m, "regression", bv, cv))
207             elif cv > bv + tols.quality_tol:
208                 res.findings.append(Finding(v, m, "improvement", bv, cv))
209         return res
210
211
212     # ----- #
213     # Report formatting (pure).
214     # ----- #
215     def _q(qd: Optional[Dict[str, Any]], key: str) -> str:
216         if not qd or qd.get(key) is None:
217             return "?"
218         return f"{qd[key]:.3f}"
219
220
221     def format_report(baseline: Dict[str, Any], current: Dict[str, Any],
222                     result: NightlyResult, date: str, head: str) -> str:
223         status = result.overall_status()
224         badge = {"PASS": "? PASS", "REGRESSION": "? REGRESSION",
225                 "STALE": "? STALE (re-freeze the baseline)"}[status]
226         cost = current.get("cost", {})
227         L: List[str] = [
228             f"# Nightly engine regression (reel-count + quality + cost) ? {date}",
229             "",
230             f"***Status: {badge}** · exit code {result.exit_code()} · segmenter
`{current.get('segmenter')}`",
231             "",
232             f"- HEAD `{head}` · baseline `{baseline.get('git_commit', '?')}` (frozen
{baseline.get('generated_at', '?')})",
233             f"- **Run cost: ${cost.get('usd', 0):.4f} ({cost.get('inr', 0):.2f})** · "
234             f"billed {cost.get('billed_seconds', 0):.0f}s @ ${cost.get('machine_usd_per_hr',
0)}/hr",
235             "",
236         ]
237         if result.regressions:
238             L += ["## ? Regressions", ""] + [f"- {f.message()}" for f in result.regressions]
239             + [""]
240         if result.stale:
241             L += ["## ? Stale / not comparable", ""] + [f"- {f.message()}" for f in
result.stale]
242         L += ["- Action: `python scripts/nightly_reelcount.py --update-baseline` (after
confirming the change is intended).", ""]
243
244         # Per-video table.
245         b_pv, c_pv = baseline.get("per_video", {}), current.get("per_video", {})
246         L += ["## Per-video", ""]
247         "| video | reels (base?now) | tol_f1 (base?now) | f1@0.5 | status |",
248         "|---|---|---|---|---|",

```

```

248     for v in sorted(set(b_pv) | set(c_pv)):
249         b, c = b_pv.get(v, {}), c_pv.get(v, {})
250         bc = b.get("reel_count")
251         cc = c.get("reel_count")
252         rc = f"'?' if bc is None else bc}{'ERROR' if not c.get('ok', True) else ('?'
if cc is None else cc})"
253         bq, cq = b.get("quality"), c.get("quality")
254         tf = f"_{q(bq, 'tol_f1')}?_{q(cq, 'tol_f1')}"
255         f5 = f"_{q(bq, 'f1@0.5')}?_{q(cq, 'f1@0.5')}"
256         vstatus = "?" if any(f.video == v and f.kind == "regression" for f in
result.findings) else "?"
257         L.append(f"| {v} | {rc} | {tf} | {f5} | {vstatus} |")
258     L.append("")
259
260     if result.improvements:
261         L += ["_Improvements over baseline (re-freeze to ratchet up):_"] \
262             + [f"- {f.message()}" for f in result.improvements] + [""]
263     L += ["---",
264         "_Full configured+checkpointed pipeline on the real model (GCP GPU VM).
Tripwire only ? does "
265         "not change the promotion gate. Complements PR #202's cached-trajectory CPU
re-score._"]
266     return "\n".join(L)
267
268
269     # ----- #
270     # IO shell ? running the configured pipeline per video (needs backend + GPU + video).
271     # ----- #
272     def _load_gts(labels_ref: Optional[str]) -> Optional[List[Tuple[float, float]]]:
273         """Load ground-truth rally intervals from a golden-features JSON (`gts` key) or a
rallies CSV
274         (`start,end` columns / pairs). Returns None when no labels are configured."""
275         if not labels_ref:
276             return None
277         from backend.storage import fetch
278         local = fetch(labels_ref)
279         if local.endswith(".json"):
280             with open(local, encoding="utf-8") as fh:
281                 data = json.load(fh)
282                 return [(float(s), float(e)) for s, e in (data.get("gts") or [])]
283         # CSV: header start,end (or first two numeric columns)
284         import csv
285         out: List[Tuple[float, float]] = []
286         with open(local, encoding="utf-8") as fh:
287             for row in csv.reader(fh):
288                 try:
289                     out.append((float(row[0]), float(row[1])))
290                 except (ValueError, IndexError):
291                     continue
292         return out
293
294
295     def _segments_to_intervals(segs: List[Dict[str, Any]]) -> List[Tuple[float, float]]:
296         """(`start, end` pairs from segmenter result dicts, tolerant of the two key
conventions in the
297         codebase (`start_time`/`end_time` ? motion/DB shape ? or `start`/`end`).
Used only for the
298         optional quality metrics; the reel COUNT is `len(segs)` regardless, so a key
mismatch degrades
299         quality scoring gracefully (skipped) without ever miscounting reels."""
300         out: List[Tuple[float, float]] = []
301         for s in segs:
302             start = s.get("start_time", s.get("start"))
303             end = s.get("end_time", s.get("end"))
304             if start is not None and end is not None:

```

```

305         out.append((float(start), float(end)))
306     return out
307
308
309     def run_one_video(video: Dict[str, Any], segmenter: str, config: Any,
310                      rerun_on_mismatch: bool = True,
311                      baseline_count: Optional[int] = None) -> Dict[str, Any]:
312         """Run the configured pipeline on one video ? reel count (+ quality vs labels). Lazy
backend imports.
313
314         The re-run-once guard (the one known non-determinism ? a transient
``add_play_segment``?None drop,
315         database.py): if the count != the frozen baseline, re-process the video ONCE; a
transient drop won't
316         reproduce, a real change will."""
317         import random
318         import time
319
320         from backend.eval import rally_seg_eval
321         from backend.infrastructure.database import Database
322         from backend.pipeline.segmenters import segmenter_registry
323         from backend.results import Failure
324         from backend.storage import fetch
325         from backend.utils.hashing import compute_video_id
326
327         name = video["name"]
328         try:
329             local = fetch(video["uri"])
330         except Exception as e: # noqa: BLE001 ? surface, never hide
331             return {"name": name, "reel_count": None, "ok": False, "error": f"fetch failed:
{e}",
332                    "quality": None, "wall_seconds": 0.0}
333
334         gts = _load_gts(video.get("labels_uri"))
335         seg_cls = segmenter_registry.get(segmenter)
336         if not seg_cls:
337             return {"name": name, "reel_count": None, "ok": False,
338                    "error": f"segmenter '{segmenter}' not registered", "quality": None,
"wall_seconds": 0.0}
339
340         def _process(db_path: str) -> Tuple[Optional[int], Optional[List[Tuple[float,
float]]], float, Optional[str]]:
341             random.seed(0) # close motion.py's unseeded metadata RNG so secondary fields
are reproducible too
342             db = Database(db_path)
343             video_id = compute_video_id(local)
344             t0 = time.monotonic()
345             result = seg_cls(db, config).process_video(video_id, local)
346             wall = time.monotonic() - t0
347             if isinstance(result, Failure):
348                 return None, None, wall, getattr(result, "message", "segmenter failed")
349             segs = result.value if hasattr(result, "value") else result
350             intervals = _segments_to_intervals(segs)
351             return len(segs), intervals, wall, None
352
353         import tempfile
354         total_wall = 0.0
355         with tempfile.TemporaryDirectory(prefix="nightly_reel_") as td:
356             count, intervals, wall, err = _process(os.path.join(td, "run1.db"))
357             total_wall += wall
358             if err is not None:
359                 return {"name": name, "reel_count": None, "ok": False, "error": err,
360                        "quality": None, "wall_seconds": round(total_wall, 2)}
361             # re-run-once guard for the transient DB-drop flake
362             if rerun_on_mismatch and baseline_count is not None and count != baseline_count:

```

```

363         count2, intervals2, wall2, err2 = _process(os.path.join(td, "run2.db"))
364         total_wall += wall2
365         if err2 is None and count2 is not None:
366             count, intervals = count2, intervals2 # the reproduced (or corrected)
value
367
368         quality = None
369         if gts and intervals is not None:
370             row = rally_seg_eval.score_intervals(intervals, gts)
371             quality = {k: row[k] for k in ("tol_f1", "f1@0.5", "f1@0.25",
"tol_precision", "tol_recall")}
372             if k in row}
373
374         return {"name": name, "reel_count": count, "ok": True, "error": None,
375             "quality": quality, "wall_seconds": round(total_wall, 2)}
376
377
378     def load_manifest(path: str) -> Dict[str, Any]:
379         with open(path, encoding="utf-8") as fh:
380             return json.load(fh)
381
382
383     def run_corpus(manifest: Dict[str, Any], baseline: Optional[Dict[str, Any]],
384         rerun_on_mismatch: bool = True) -> Tuple[List[Dict[str, Any]], float]:
385         """Run every manifest video through the configured pipeline. Returns (run_results,
total_wall_s)."""
386         from backend.config import load_config
387
388         config = load_config()
389         # apply manifest config overrides onto the typed config (dotted keys)
390         for dotted, val in (manifest.get("config_overrides") or {}).items():
391             _set_dotted(config, dotted, val)
392         segmenter = manifest.get("segmenter") or getattr(getattr(config, "indexing", None),
"default_segmenter", "motion")
393         b_pv = (baseline or {}).get("per_video", {})
394         results: List[Dict[str, Any]] = []
395         total_wall = 0.0
396         for v in manifest.get("videos", []):
397             base_count = (b_pv.get(v["name"]) or {}).get("reel_count")
398             r = run_one_video(v, segmenter, config, rerun_on_mismatch, base_count)
399             total_wall += float(r.get("wall_seconds") or 0.0)
400             results.append(r)
401         return results, total_wall
402
403
404     def _set_dotted(config: Any, dotted: str, value: Any) -> None:
405         """Best-effort dotted-path set onto a pydantic config (e.g.
'indexing.motion_threshold')."""
406         parts = dotted.split(".")
407         obj = config
408         for p in parts[:-1]:
409             obj = getattr(obj, p, None)
410             if obj is None:
411                 return
412         try:
413             setattr(obj, parts[-1], value)
414         except Exception: # noqa: BLE001 ? overrides are best-effort; a bad key shouldn't
crash the run
415             pass
416
417
418     def _git_head() -> str:
419         import subprocess
420         try:
421             out = subprocess.run(["git", "-C", REPO_ROOT, "rev-parse", "--short", "HEAD"],

```

```

422             capture_output=True, text=True, timeout=10)
423         return out.stdout.strip() or "?"
424     except Exception: # noqa: BLE001
425         return "?"
426
427
428     def load_baseline(path: str) -> Optional[Dict[str, Any]]:
429         if not os.path.exists(path):
430             return None
431         with open(path, encoding="utf-8") as fh:
432             return json.load(fh)
433
434
435     def save_baseline(path: str, snapshot: Dict[str, Any]) -> None:
436         out = dict(snapshot)
437         out["generated_at"] = _dt.date.today().isoformat()
438         out["git_commit"] = _git_head()
439         out["_doc"] = ("Frozen nightly reel-count baseline. Regenerate with `python
scripts/nightly_reelcount.py "
440             "--update-baseline` after an INTENDED engine change or when adding a
video. Tied to the "
441             "configured+checkpointed engine + the manifest corpus.")
442         out.pop("cost", None) # cost is per-run, not part of the frozen baseline
443         os.makedirs(os.path.dirname(path), exist_ok=True)
444         tmp = path + ".tmp"
445         with open(tmp, "w", encoding="utf-8") as fh:
446             json.dump(out, fh, indent=2, sort_keys=True)
447             fh.write("\n")
448         os.replace(tmp, path)
449
450
451     def build_parser() -> argparse.ArgumentParser:
452         p = argparse.ArgumentParser(description="Nightly E2E engine regression: reel-count +
quality + cost.")
453         p.add_argument("--manifest", default=DEFAULT_MANIFEST)
454         p.add_argument("--baseline", default=DEFAULT_BASELINE)
455         p.add_argument("--report-dir", default=DEFAULT_REPORT_DIR)
456         p.add_argument("--quality-tol", type=float, default=DEFAULT_QUALITY_TOL)
457         p.add_argument("--vm-uptime-seconds", type=float, default=None,
458             help="authoritative VM uptime (boot?delete) for the cost line; "
459             "defaults to the in-process pipeline wall time (a lower bound)")
460         p.add_argument("--no-rerun-guard", action="store_true",
461             help="disable the re-run-once-on-mismatch guard (the DB-drop flake
mitigation)")
462         p.add_argument("--update-baseline", action="store_true",
463             help="freeze the current reel counts + quality as the new baseline
instead of checking")
464         p.add_argument("--email", action="store_true", help="email the report (see
scripts/nightly_email.py)")
465         p.add_argument("--email-to", default=None, help="override the report recipient")
466         p.add_argument("--date", default=None, help="override the report date (YYYY-MM-DD;
for tests)")
467         p.add_argument("--no-report", action="store_true", help="skip writing the dated
report file")
468         return p
469
470
471     def main(argv: Optional[List[str]] = None) -> int:
472         import logging
473         logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")
474         log = logging.getLogger("nightly_reelcount")
475         args = build_parser().parse_args(argv)
476         tols = Tolerances(quality_tol=args.quality_tol)
477         date = args.date or _dt.date.today().isoformat()
478

```

```

479         if not os.path.exists(args.manifest):
480             log.error("manifest not found: %s", args.manifest)
481             return 2
482         manifest = load_manifest(args.manifest)
483         baseline = load_baseline(args.baseline)
484
485         try:
486             run_results, total_wall = run_corpus(manifest, baseline, rerun_on_mismatch=not
args.no_rerun_guard)
487         except Exception as e: # noqa: BLE001 ? operational failure (deps/config) ? exit 2,
never a false "pass"
488             log.error("run failed: %s", e)
489             return 2
490         if not run_results:
491             log.error("no videos in manifest %s", args.manifest)
492             return 2
493
494         billed = args.vm_uptime_seconds if args.vm_uptime_seconds is not None else
total_wall
495         cost = cost_report(
496             billed_seconds=billed,
497             machine_usd_per_hr=float(manifest.get("machine_usd_per_hr",
DEFAULT_MACHINE_USD_PER_HR)),
498             fx_inr_per_usd=float(manifest.get("fx_inr_per_usd", DEFAULT_FX_INR_PER_USD)),
499             gpu_seconds=total_wall,
500         )
501         segmenter = manifest.get("segmenter", "motion")
502         snapshot = summarize_results(run_results, segmenter, cost)
503         snapshot["git_commit"] = _git_head()
504         snapshot["generated_at"] = date
505
506         if args.update_baseline:
507             save_baseline(args.baseline, snapshot)
508             log.info("baseline frozen ? %s", args.baseline)
509             for r in run_results:
510                 log.info(" %s: reels=%s ok=%s", r["name"], r.get("reel_count"),
r.get("ok"))
511             return 0
512
513         if baseline is None:
514             log.error("no baseline at %s ? freeze one with --update-baseline (on the
configured engine).", args.baseline)
515             return 3
516
517         result = compare(baseline, snapshot, tols)
518         report = format_report(baseline, snapshot, result, date, snapshot["git_commit"])
519
520         if not args.no_report:
521             os.makedirs(args.report_dir, exist_ok=True)
522             md_path = os.path.join(args.report_dir, f"{date}.md")
523             with open(md_path, "w", encoding="utf-8") as fh:
524                 fh.write(report + "\n")
525             with open(os.path.join(args.report_dir, f"{date}.json"), "w", encoding="utf-8")
as fh:
526                 json.dump({"date": date, "status": result.overall_status(), "exit_code":
result.exit_code(),
527                             "snapshot": snapshot, "regressions": [f.message() for f in
result.regressions]},
528                             fh, indent=2, sort_keys=True)
529             log.info("report ? %s", md_path)
530
531         print(report)
532
533         if args.email:
534             try:

```

```

535         from scripts.nightly_email import send_report
536         subject = f"[KhelSutra nightly] {result.overall_status()} ? reel-count
engine regression {date}"
537         sent = send_report(subject, report, to=args.email_to)
538         log.info("email %s", "sent" if sent else "skipped (no creds ? report still
written)")
539         except Exception as e: # noqa: BLE001 ? a broken mailer must NOT mask the
regression verdict
540             log.error("email failed (report still written): %s", e)
541
542         code = result.exit_code()
543         if code == 1:
544             log.error("REEL-COUNT REGRESSION (%d) ? see %s", len(result.regressions),
args.report_dir)
545         elif code == 4:
546             log.warning("baseline STALE ? corpus/segmenter changed; re-freeze with --update-
baseline")
547         else:
548             log.info("nightly reel-count check PASSED (cost $%.4f / ?%.2f)", cost["usd"],
cost["inr"])
549         return code
550
551
552     if __name__ == "__main__":
553         raise SystemExit(main())
554

```

## 6. user (2026-06-22T09:47:26.987Z)

```

1     #!/usr/bin/env bash
2     # Launch ONE nightly engine-regression run on an ephemeral GCP GPU VM, then return. The
VM runs
3     # run_on_vm.sh as its startup-script (regression ? email ? SELF-DELETE), so this just
stages the repo
4     # and kicks the VM off. Driver-agnostic: call it from Cloud Scheduler?Cloud Run job, a
cron on any
5     # always-on box, the owner's machine, or a Claude routine. Needs gcloud auth + the repo
checkout.
6     #
7     #   bash deploy/gcp/nightly_regression/launch.sh                               # L4, stages HEAD,
kicks the VM
8     #   RALLY_GPU=t4 RALLY_REF=$(git rev-parse HEAD) bash
deploy/gcp/nightly_regression/launch.sh
9     #
10    # ? Run this MANUALLY once as a dry-run (watch the serial log) before wiring a scheduler
? the VM-side
11    # flow is NOT covered by CI. Tail: gcloud compute instances get-serial-port-output
<name> --zone=<z>.
12    set -uo pipefail
13
14    PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
15    BUCKET="${RALLY_BUCKET:-kheIsutra-rally-corpus}"
16    SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"
17    GPU="${RALLY_GPU:-l4}"
18    NAME="${RALLY_VM_NAME:-rally-nightly-$(date -u +%Y%m%d)}"
19    REF="${RALLY_REF:-HEAD}"
20    WEIGHTS_URI="${RALLY_WEIGHTS_URI:-gs://$BUCKET/weights/wasb_badminton_best.pth.tar}"
21    REPORT_PREFIX="${RALLY_REPORT_PREFIX:-nightly/reports}"
22    DO_EMAIL="${RALLY_EMAIL:-true}"
23    SMTP_SECRET="${NIGHTLY_SMTP_SECRET:-projects/$PROJECT/secrets/nightly-smtp-
pass/versions/latest}"
24    SMTP_USER="${NIGHTLY_SMTP_USER:-}"
25    EMAIL_TO="${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
26    HERE="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
27    STARTUP="$HERE/run_on_vm.sh"

```

```

28     export CLOUDSDK_CORE_DISABLE_PROMPTS=1
29
30     ZONES="$(RALLY_GCP_ZONES:-asia-south1-a asia-south1-b asia-south1-c asia-southeast1-a
asia-southeast1-b asia-southeast1-c)"
31     if [ "$GPU" = "l4" ]; then MACHINE="g2-standard-4"; ACCEL=(); else
MACHINE="n1-standard-8"; ACCEL=(--accelerator=type=nvidia-tesla-t4,count=1); fi
32
33     # 1) Stage the repo @ REF to GCS (no GitHub auth on the VM; the worker SA can read the
bucket).
34     SHA="$(git rev-parse "$REF")"
35     TGZ_URI="gs://$BUCKET/nightly/repo/${SHA}.tgz"
36     echo "== staging repo $SHA -> $TGZ_URI =="
37     TMP_TGZ="$(mktemp --suffix=.tgz)"
38     git archive --format=tar.gz -o "$TMP_TGZ" "$REF"
39     gcloud storage cp "$TMP_TGZ" "$TGZ_URI" --project="$PROJECT"
40     rm -f "$TMP_TGZ"
41
42     # 2) Zone-sweep create the VM (mirrors create_gpu_vm.sh) with run_on_vm.sh + the config
metadata.
43     ERR="$(mktemp)"
44     for Z in $ZONES; do
45         echo "== creating $NAME ($GPU) in $Z =="
46         if gcloud compute instances create "$NAME" --project="$PROJECT" --zone="$Z" \
47             --machine-type="$MACHINE" "${ACCEL[@]}" \
48             --maintenance-policy=TERMINATE --provisioning-model=STANDARD \
49             --image-family=common-cu129-ubuntu-2204-nvidia-580 --image-project=deeplearning-
platform-release \
50             --boot-disk-size=120GB --boot-disk-type=pd-balanced \
51             --service-account="$SA" --scopes=cloud-platform \
52             --metadata-from-file=startup-script="$STARTUP" \
53             --metadata=repo-tgz-uri="$TGZ_URI",git-sha="$SHA",gcs-bucket="$BUCKET",report-
prefix="$REPORT_PREFIX",weights-uri="$WEIGHTS_URI",email="$DO_EMAIL",smtp-
secret="$SMTP_SECRET",smtp-user="$SMTP_USER",email-to="$EMAIL_TO" \
54             2>"$ERR"; then
55             echo "? launched $NAME in $Z ? it will run the regression, email $EMAIL_TO, and
self-delete."
56             echo "    serial log: gcloud compute instances get-serial-port-output $NAME --zone=$Z
--project=$PROJECT"
57             echo "    if it doesn't self-delete: gcloud compute instances delete $NAME --zone=$Z
-q"
58             rm -f "$ERR"; exit 0
59         fi
60         if grep -qiE "STOCKOUT|does not have enough" "$ERR"; then echo "    (no $GPU capacity in
$Z ? next zone)"; continue; fi
61         echo "    create failed:"; tail -4 "$ERR"; rm -f "$ERR"; exit 1
62     done
63     echo "? no $GPU capacity in any swept zone ($ZONES). Retry later or RALLY_GPU=t4."
64     rm -f "$ERR"; exit 1
65

```

## 7. assistant (2026-06-22T09:47:29.415Z)

Now let me read the remaining files to complete the review:

## 8. user (2026-06-22T09:47:30.959Z)

```

1     #!/usr/bin/env bash
2     # Wire the nightly engine-regression as a CLOUD-NATIVE, scale-to-zero job:
3     #   Cloud Scheduler (cron) ? Cloud Run JOB (the launcher image) ? launch.sh
4     #   ? ephemeral GPU VM (regression + email + SELF-DELETE).
5     # The only always-on cost is $0 (Cloud Run jobs + Scheduler are scale-to-zero; the GPU
VM is ephemeral).
6     #
7     # ? OWNER-RUN, ONCE, after a successful manual `bash
deploy/gcp/nightly_regression/launch.sh` dry-run.

```



```

8      # NOT verified in CI (no GCP/GPU here). Prereqs: GPU quota (README §2), the golden clips
+ weights in
9      # gs://<bucket>, and the SMTP secret (see README "Enable").
10     #
11     #   bash deploy/gcp/nightly_regression/register_scheduler.sh           # build + deploy
+ schedule (03:00 IST)
12     #   RALLY_CRON='0 3 * * *' RALLY_TZ='Asia/Kolkata' bash ../register_scheduler.sh
13     set -euo pipefail
14
15     PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
16     REGION="${RALLY_RUN_REGION:-asia-south1}"
17     JOB="${RALLY_RUN_JOB:-nightly_regression-launcher}"
18     SCHED="${RALLY_SCHED_NAME:-nightly_regression}"
19     CRON="${RALLY_CRON:-0 3 * * *}"          # 03:00 daily
20     TZ="${RALLY_TZ:-Asia/Kolkata}"
21     SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"
22     IMAGE="${RALLY_LAUNCHER_IMAGE:-${REGION}-docker.pkg.dev/${PROJECT}/rally/nightly-
launcher:latest}"
23     REPO_ROOT="$(git rev-parse --show-toplevel)"
24     export CLOUDSDK_CORE_DISABLE_PROMPTS=1
25
26     echo "== enable APIs (run, scheduler, build, artifact registry) =="
27     gcloud services enable run.googleapis.com cloudscheduler.googleapis.com \
28         cloudbuild.googleapis.com artifactregistry.googleapis.com --project="$PROJECT"
29
30     echo "== ensure the Artifact Registry repo 'rally' exists =="
31     gcloud artifacts repositories describe rally --location="$REGION" --project="$PROJECT"
>/dev/null 2>&1 || \
32     gcloud artifacts repositories create rally --repository-format=docker
--location="$REGION" --project="$PROJECT"
33
34     echo "== build + push the launcher image from the repo root =="
35     ( cd "$REPO_ROOT" && gcloud builds submit --project="$PROJECT" --tag "$IMAGE" \
36         --gcs-source-staging-dir="gs://${RALLY_BUILD_BUCKET:-khelsutra-rally-
corpus}/cloudbuild" \
37         -f deploy/gcp/nightly_regression/Dockerfile.launcher . )
38
39     echo "== create/update the Cloud Run JOB (launcher needs compute-admin to create the GPU
VM) =="
40     gcloud run jobs deploy "$JOB" --project="$PROJECT" --region="$REGION" \
41         --image="$IMAGE" --service-account="$SA" --max-retries=0 --task-timeout=900s \
42         --set-env-vars="RALLY_GPU=${RALLY_GPU:-
l4},RALLY_EMAIL=true,NIGHTLY_EMAIL_TO=${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
43
44     echo "== schedule it (Cloud Scheduler ? Run job admin API) =="
45     RUN_URI="https://${REGION}-
run.googleapis.com/apis/run.googleapis.com/v1/namespaces/${PROJECT}/jobs/${JOB}:run"
46     gcloud scheduler jobs describe "$SCHED" --location="$REGION" --project="$PROJECT"
>/dev/null 2>&1 && \
47     gcloud scheduler jobs delete "$SCHED" --location="$REGION" --project="$PROJECT"
--quiet || true
48     gcloud scheduler jobs create http "$SCHED" --project="$PROJECT" --location="$REGION" \
49         --schedule="$CRON" --time-zone="$TZ" \
50         --uri="$RUN_URI" --http-method=POST \
51         --oauth-service-account-email="$SA"
52
53     cat <<EOF
54
55     ? Scheduled '$SCHED' ($CRON $TZ) ? Cloud Run job '$JOB' ? launch.sh ? ephemeral GPU VM.
56     IAM the launcher SA needs (grant once if missing):
57         roles/compute.instanceAdmin.v1 (create/delete the GPU VM)
58         roles/iam.serviceAccountUser (attach $SA to the VM)
59         roles/run.invoker + roles/cloudscheduler.jobRunner (Scheduler ? Run)
60     Trigger now to test: gcloud run jobs execute $JOB --region=$REGION
--project=$PROJECT

```

```
61      Remove:          gcloud scheduler jobs delete $SCHED --location=$REGION; gcloud
run jobs delete $JOB --region=$REGION
62      EOF
63
```

#### 9. user (2026-06-22T09:47:31.614Z)

```
1      # Tiny launcher image for the nightly engine-regression Cloud Run JOB. It is NOT the GPU
workload ?
2      # it only stages the repo + creates the ephemeral GPU VM (which self-deletes). Scale-to-
zero: the job
3      # runs for a few seconds nightly, then exits. Base = google/cloud-sdk (gcloud + git
preinstalled).
4      FROM gcr.io/google.com/cloudsdktool/cloud-sdk:slim
5      WORKDIR /app
6      # The repo is the build context (run `gcloud builds submit` from the repo root, see
register_scheduler.sh).
7      COPY . /app
8      ENTRYPOINT ["bash", "deploy/gcp/nightly_regression/launch.sh"]
9
```

#### 10. user (2026-06-22T09:47:32.466Z)

```
1      {
2      "doc": "Nightly E2E engine reel-count manifest (scripts/nightly_reelcount.py). The
configured+checkpointed pipeline runs on each video on a GCP GPU VM; the reel count + quality
metrics are asserted vs eval_baselines/reelcount_baseline.json. ADD A VIDEO = append one row +
re-freeze: `python scripts/nightly_reelcount.py --update-baseline` (commit the updated
baseline). URIs may be gs:// (recommended, co-located with the VM), gdrive://, or local paths.
labels_uri (optional) is a golden-features JSON (gts key) or a start,end CSV ? present =>
quality metrics (tol_f1/f1@0.5/f1@0.25) are scored too. EDIT the example rows below to your
actual short (~5-6 min) OWNED, minors-free clips before enabling.",
3      "segmenter": "wasb_hybrid",
4      "detector_impl": "native",
5      "config_overrides": {
6      "indexing.default_segmenter": "wasb_hybrid",
7      "indexing.detector_impl": "native"
8      },
9      "machine_usd_per_hr": 0.70,
10     "fx_inr_per_usd": 83.0,
11     "videos": [
12     {
13         "name": "GX030094",
14         "uri": "gs://khehsutra-rally-corpus/nightly/videos/GX030094.mp4",
15         "labels_uri": "gs://khehsutra-rally-
corpus/nightly/labels/GX030094_golden_features.json"
16     },
17     {
18         "name": "Badminton_BXH_2",
19         "uri": "gs://khehsutra-rally-corpus/nightly/videos/Badminton_BXH_2.mp4",
20         "labels_uri": "gs://khehsutra-rally-
corpus/nightly/labels/Badminton_BXH_2_golden_features.json"
21     },
22     {
23         "name": "GX010128",
24         "uri": "gs://khehsutra-rally-corpus/nightly/videos/GX010128.mp4",
25         "labels_uri": "gs://khehsutra-rally-
corpus/nightly/labels/GX010128_golden_features.json"
26     }
27     ]
28     }
29
```

#### 11. assistant (2026-06-22T09:47:35.322Z)

Now let me search for specific code patterns around the SMTP handling and exception behavior:

## 12. user (2026-06-22T09:47:37.384Z)

```
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-1-""Email the nightly regression
report ? stdlib SMTP, creds from env or GCP Secret Manager.
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-2-
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-3-Used by
`scripts\nightly_reelcount.py --email` (and reusable by any nightly). Deliberately
scripts\nightly_email.py:4:dependency-light (stdlib `smtplib` + `email.message`) and
**graceful**: if no SMTP password can
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-5-be resolved it logs + returns
`False` (the report is still written/printed ? a broken mailer must
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-6-never mask a regression).
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-7-
--
scripts\nightly_email.py-55-         from google.cloud import secretmanager # type: ignore
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-56-         client =
secretmanager.SecretManagerServiceClient()
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-57-         return
client.access_secret_version(name=secret_ref).payload.data.decode("utf-8")
scripts\nightly_email.py:58:     except Exception: # noqa: BLE001 ? lib absent or auth issue ?
try the CLI
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-59-         pass
scripts\nightly_email.py-60-     try:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-61-         import subprocess
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-63-         out =
subprocess.run(["gcloud", "secrets", "versions", "access", "latest", "--secret", name],
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-64-         capture_output=True, text=True, timeout=30)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-65-         return
out.stdout.strip() if out.returncode == 0 and out.stdout.strip() else None
scripts\nightly_email.py:66:     except Exception as e: # noqa: BLE001
scripts\nightly_email.py-67-         log.warning("secret-manager resolution failed: %s", e)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-68-         return None
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-69-
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-81-         log.warning("no SMTP
password (set NIGHTLY_SMTP_PASS or NIGHTLY_SMTP_SECRET) ? email skipped.")
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-82-         return False
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-83-
scripts\nightly_email.py:84:     import smtplib
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-85-     msg =
build_message(subject, body_md, from_addr=user, to_addr=to_addr)
scripts\nightly_email.py:86:     with smtplib.SMTP(host, port, timeout=60) as s:
```

```

C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-87-        s.starttls()
scripts\nightly_email.py:88:            s.login(user, password)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-89-        s.send_message(msg)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-90-        log.info("emailed report
to %s", to_addr)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_email.py-91-        return True
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-327-        name = video["name"]
scripts\nightly_reelcount.py-328-        try:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-329-            local =
fetch(video["uri"])
scripts\nightly_reelcount.py:330:        except Exception as e: # noqa: BLE001 ? surface, never
hide
scripts\nightly_reelcount.py-331-            return {"name": name, "reel_count": None, "ok": False,
"error": f"fetch failed: {e}",
scripts\nightly_reelcount.py-332-                "quality": None, "wall_seconds": 0.0}
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-333-
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-411-            return
scripts\nightly_reelcount.py-412-        try:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-413-            setattr(obj,
parts[-1], value)
scripts\nightly_reelcount.py:414:        except Exception: # noqa: BLE001 ? overrides are best-
effort; a bad key shouldn't crash the run
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-415-            pass
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-416-
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-417-
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-421-        out =
subprocess.run(["git", "-C", REPO_ROOT, "rev-parse", "--short", "HEAD"],
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-422-        capture_output=True, text=True, timeout=10)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-423-        return
out.stdout.strip() or "?"
scripts\nightly_reelcount.py:424:        except Exception: # noqa: BLE001
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-425-        return "?"
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-426-
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-427-
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-484-
scripts\nightly_reelcount.py-485-        try:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-486-            run_results,
total_wall = run_corpus(manifest, baseline, rerun_on_mismatch=not args.no_rerun_guard)
scripts\nightly_reelcount.py:487:        except Exception as e: # noqa: BLE001 ? operational
failure (deps/config) ? exit 2, never a false "pass"

```

```

scripts\nightly_reelcount.py-488-         log.error("run failed: %s", e)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-489-         return 2
scripts\nightly_reelcount.py-490-         if not run_results:
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-536-         subject =
f"[KhelSutra nightly] {result.overall_status()} ? reel-count engine regression {date}"
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-537-         sent =
send_report(subject, report, to=args.email_to)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-538-         log.info("email %s", "sent" if sent else "skipped (no creds ? report still written)")
scripts\nightly_reelcount.py:539:         except Exception as e: # noqa: BLE001 ? a broken
mailer must NOT mask the regression verdict
scripts\nightly_reelcount.py-540-         log.error("email failed (report still written):
%s", e)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-541-
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_reelcount.py-542-         code =
result.exit_code()
--
scripts\doctor.py-51-     try:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\doctor.py-52-         from backend.config import
save_default_config
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\doctor.py-53-         from backend.utils.app_home
import default_config_path
scripts\doctor.py:54:     except Exception as e:
scripts\doctor.py-55-         print(f"[ERROR] Could not import backend.config loader: {e}")
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\doctor.py-56-         print("[INFO] Falling back to
no-op for config initialization.")
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\doctor.py-57-         return
--
scripts\doctor.py-67-     try:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\doctor.py-68-         saved_path =
save_default_config(config_path)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\doctor.py-69-         print(f"[SUCCESS] config.json
initialized with default settings at {saved_path}.")
scripts\doctor.py:70:     except Exception as e:
scripts\doctor.py-71-         print(f"[ERROR] Failed to create config.json via loader: {e}")
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\doctor.py-72-
scripts\doctor.py-73-def run_diagnostics():
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_regression.py-469-         out =
subprocess.run(["git", "-C", REPO_ROOT, "rev-parse", "--short", "HEAD"],
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_regression.py-470-         capture_output=True, text=True, timeout=10)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_regression.py-471-         return
out.stdout.strip() or "?"
scripts\nightly_regression.py:472:     except Exception:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_regression.py-473-         return "?"
C:\Users\avidu\Projects\badminton-highlight-

```

```

indexer\.claude\worktrees\nightly-e2e\scripts\nightly_regression.py-474-
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\nightly_regression.py-475-
--
scripts\run_process_and_stitch.py-132-            if isinstance(meta, str):
scripts\run_process_and_stitch.py-133-                try:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\run_process_and_stitch.py-134-                    meta
= json.loads(meta)
scripts\run_process_and_stitch.py:135:                except Exception:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\run_process_and_stitch.py-136-                    meta
= {}
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\run_process_and_stitch.py-137-                    shot_count =
meta.get("shot_count", 0)
scripts\run_process_and_stitch.py-138-                    if shot_count >= min_shot_count:
--
scripts\run_process_and_stitch.py-160-            try:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\run_process_and_stitch.py-161-                out_path =
compiler.compile_highlights(video_path, time_pairs, output_filename)
scripts\run_process_and_stitch.py-162-                print(f"Stitched highlight video saved to:
{out_path}", flush=True)
scripts\run_process_and_stitch.py:163:            except Exception as e:
scripts\run_process_and_stitch.py-164-                print(f"Error during stitching: {e}", flush=True)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\run_process_and_stitch.py-165-
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e\scripts\run_process_and_stitch.py-166-
_log_rally_summary()

```

### 13. assistant (2026-06-22T09:47:40.543Z)

Now let me examine the exception handling in the context where `send\_report` is called:

### 14. user (2026-06-22T09:47:41.674Z)

```

530
531     print(report)
532
533     if args.email:
534         try:
535             from scripts.nightly_email import send_report
536             subject = f"[KhelSutra nightly] {result.overall_status()} ? reel-count
engine regression {date}"
537             sent = send_report(subject, report, to=args.email_to)
538             log.info("email %s", "sent" if sent else "skipped (no creds ? report still
written)")
539             except Exception as e: # noqa: BLE001 ? a broken mailer must NOT mask the
regression verdict
540                 log.error("email failed (report still written): %s", e)
541
542     code = result.exit_code()
543     if code == 1:
544         log.error("REEL-COUNT REGRESSION (%d) ? see %s", len(result.regressions),
args.report_dir)

```

### 15. assistant (2026-06-22T09:47:43.806Z)

Now let me verify the exact lines of the SMTP block to confirm the finding:

### 16. user (2026-06-22T09:47:44.869Z)

```

71     def send_report(subject: str, body_md: str, to: Optional[str] = None,
72                     env: Optional[dict] = None) -> bool:
73         """Send the report. Returns True if sent, False if skipped (no creds) ? never raises
on missing creds."""
74         env = env if env is not None else os.environ
75         to_addr = to or env.get("NIGHTLY_EMAIL_TO") or DEFAULT_TO
76         host = env.get("NIGHTLY_SMTP_HOST", DEFAULT_HOST)
77         port = int(env.get("NIGHTLY_SMTP_PORT", DEFAULT_PORT))
78         user = env.get("NIGHTLY_SMTP_USER") or to_addr
79         password = resolve_secret(env)
80         if not password:
81             log.warning("no SMTP password (set NIGHTLY_SMTP_PASS or NIGHTLY_SMTP_SECRET) ?
email skipped.")
82             return False
83
84         import smtplib
85         msg = build_message(subject, body_md, from_addr=user, to_addr=to_addr)
86         with smtplib.SMTP(host, port, timeout=60) as s:
87             s.starttls()
88             s.login(user, password)
89             s.send_message(msg)
90         log.info("emailed report to %s", to_addr)
91         return True
92

```

#### 17. assistant (2026-06-22T09:47:49.446Z)

Now let me check if there are any log configuration details that might capture exception  
 tracebacks:

#### 18. user (2026-06-22T09:47:50.527Z)

```

scripts/nightly_reelcount.py:    logging.basicConfig(level=logging.INFO, format="%(levelname)s
%(message)s")
scripts/nightly_regression.py:    logging.basicConfig(level=logging.INFO, format="%(levelname)s
%(message)s")
scripts/run_phase3_eval.py:    logging.basicConfig(level=logging.INFO, format="%(levelname)s
%(message)s")

```

#### 19. assistant (2026-06-22T09:47:52.449Z)

Let me check where `send\_report` exceptions would propagate to and how they're logged:

#### 20. user (2026-06-22T09:48:00.421Z)

Traceback (most recent call last):

```

File "<string>", line 8, in <module>
    s.login('test@gmail.com', 'wrong_password')
~~~~~^~~~~~
File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\smtplib.py", line 764, in
login
    raise last_exception
File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\smtplib.py", line 753, in
login
    (code, resp) = self.auth(
        ~~~~~~^
        authmethod, getattr(self, method_name),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
        initial_response_ok=initial_response_ok)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\smtplib.py", line 671, in
auth
    raise SMTPAuthenticationError(code, resp)
smtplib.SMTPAuthenticationError: (535, b'5.7.8 Username and Password not accepted. For more

```

```
information, go to\n5.7.8 https://support.google.com/mail/?p=BadCredentials
d9443c01a7336-2c7436af54dsm73937305ad.11 - gsmtplib')
Exception type: SMTPAuthenticationError
Exception message: (535, b'5.7.8 Username and Password not accepted. For more information, go
to\n5.7.8 https://support.google.com/mail/?p=BadCredentials
d9443c01a7336-2c7436af54dsm73937305ad.11 - gsmtplib')
Exception args: (535, b'5.7.8 Username and Password not accepted. For more information, go
to\n5.7.8 https://support.google.com/mail/?p=BadCredentials
d9443c01a7336-2c7436af54dsm73937305ad.11 - gsmtplib')
Traceback:
```

## 21. assistant (2026-06-22T09:48:02.157Z)

Let me check what happens with different SMTP exceptions:

## 22. user (2026-06-22T09:48:14.445Z)

```
ERROR SMTP login failed
Traceback (most recent call last):
  File "<stdin>", line 34, in <module>
  File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\smtplib.py", line 764, in
login
    raise last_exception
  File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\smtplib.py", line 753, in
login
    (code, resp) = self.auth(
        ~~~~~~^
        authmethod, getattr(self, method_name),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
        initial_response_ok=initial_response_ok)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\smtplib.py", line 671, in
auth
    raise SMTPAuthenticationError(code, resp)
smtplib.SMTPAuthenticationError: (535, b'5.7.8 Username and Password not accepted. For more
information, go to\n5.7.8 https://support.google.com/mail/?p=BadCredentials
d2e1a72fcc58-84564e9e0d5sm6889803b3a.40 - gsmtplib')
=====
Test 1: SMTPAuthenticationError details
=====
Exception type: SMTPAuthenticationError
Exception repr: SMTPAuthenticationError(535, b'5.7.8 Username and Password not accepted. For
more information, go to\n5.7.8 https://support.google.com/mail/?p=BadCredentials
d9443c01a7336-2c74478ff1csm72590575ad.83 - gsmtplib')
Exception str: (535, b'5.7.8 Username and Password not accepted. For more information, go
to\n5.7.8 https://support.google.com/mail/?p=BadCredentials
d9443c01a7336-2c74478ff1csm72590575ad.83 - gsmtplib')
Exception args: (535, b'5.7.8 Username and Password not accepted. For more information, go
to\n5.7.8 https://support.google.com/mail/?p=BadCredentials
d9443c01a7336-2c74478ff1csm72590575ad.83 - gsmtplib')
'badpass' in exception: False

=====
Test 2: What log.error() captures
=====
```

## 23. assistant (2026-06-22T09:48:24.834Z)

Now let me trace the actual flow and check the context manager behavior:

## 24. user (2026-06-22T09:48:34.341Z)

```
ERROR email failed (report still written): (535, b'5.7.8 Username and Password not accepted. For
more information, go to\n5.7.8 https://support.google.com/mail/?p=BadCredentials
```



Testing send\_report flow with SMTP exception:

```
Exception message: (535, b'5.7.8 Username and Password not accepted. For more information, go
to\n5.7.8  https://support.google.com/mail/?p=BadCredentials
98e67ed59e1d1-37d1558b599sm9676239a91.7 - gsmtpt')
```

```
Checking if password is in logged output...
(above log.error output)
```

Key finding: the password is NOT in the SMTPAuthenticationError exception BUT if a caller logs the TRACEBACK with exc\_info=True, credentials in the call stack (local variables) could be exposed in the traceback.

25. assistant (2026-06-22T09:48:36.348Z)

Let me check what happens when the exception is propagated up and logged with traceback:

26. user (2026-06-22T09:48:46.724Z)

```
smtpplib.SMTPAuthenticationError: (535, b'5.7.8 Username and Password not accepted. For more
information, go to\n5.7.8 https://support.google.com/mail/?p=BadCredentials
d9443c01a7336-2c7439f8e42sm70796205ad.48 - gsmtplib')
```

541:

## 27. assistant (2026-06-22T09:48:49.329Z)

Perfect. Now let me analyze what the code ACTUALLY does vs. what the finding claims. Let me also check if exceptions from send\_report would even reach that caller:

## 28. user (2026-06-22T09:48:50.644Z)

```
71     def send_report(subject: str, body_md: str, to: Optional[str] = None,
72                     env: Optional[dict] = None) -> bool:
73         """Send the report. Returns True if sent, False if skipped (no creds) ? never raises
on missing creds."""
74         env = env if env is not None else os.environ
75         to_addr = to or env.get("NIGHTLY_EMAIL_TO") or DEFAULT_TO
76         host = env.get("NIGHTLY_SMTP_HOST", DEFAULT_HOST)
77         port = int(env.get("NIGHTLY_SMTP_PORT", DEFAULT_PORT))
78         user = env.get("NIGHTLY_SMTP_USER") or to_addr
79         password = resolve_secret(env)
80         if not password:
81             log.warning("no SMTP password (set NIGHTLY_SMTP_PASS or NIGHTLY_SMTP_SECRET) ?
email skipped.")
82             return False
83
84         import smtplib
85         msg = build_message(subject, body_md, from_addr=user, to_addr=to_addr)
86         with smtplib.SMTP(host, port, timeout=60) as s:
87             s.starttls()
88             s.login(user, password)
89             s.send_message(msg)
90         log.info("emailed report to %s", to_addr)
91         return True
92
```

## 29. assistant (2026-06-22T09:49:02.172Z)

Now I have all the evidence. Let me analyze this systematically:

**\*\*ANALYSIS OF THE FINDING:\*\***

- \*\*The claim\*\*:** "The `send_report()` function does not wrap `s.login(user, password)` in a try-except block. If SMTP login fails, Python's default exception handler will raise an `smtplib.SMTPAuthenticationError`, which may include the username and password in the exception message/traceback when logged or printed."
- \*\*What the code actually does\*\*** (lines 86-89 of `nightly_email.py`):
  - Lines 86-89 are inside a context manager (`with smtplib.SMTP(...)`)
  - There is NO try-except wrapping the login
  - If login fails, an exception will propagate up
- \*\*What happens to the exception\*\*:**
  - The exception propagates to the caller at `nightly_reelcount.py` lines 533-540
  - The caller catches it with a bare `except Exception as e:`
  - The caller logs with: `log.error("email failed (report still written): %s", e)` (line 540)
  - Note: This does NOT use `exc_info=True`, so traceback is NOT logged
- \*\*Password exposure analysis\*\*:**
  - `SMTPAuthenticationError` does NOT contain the password in its message; only the SMTP server's response
  - Local variables (like `password`) are in the stack frame but NOT in the exception object itself
  - Since the caller uses `log.error(..., e)` WITHOUT `exc_info=True`, the traceback (which would expose local variables) is NOT logged
  - The password variable exists in the frame but is not exposed unless someone uses `exc_info=True` or prints the full traceback
- \*\*Production risk\*\*:**

- The code currently handles this reasonably: exceptions are caught and logged without traceback
- However, there IS a latent issue: the `password` variable exists in the frame and could be exposed if:
  - The exception were logged with `exc\_info=True` anywhere in the orchestration
  - The exception were re-raised and the traceback were printed to console/logs
  - A debugger attached to the running process

Let me verify if there's ANY path where `exc\_info=True` is used:

### 30. user (2026-06-22T09:49:03.415Z)

No matches found

### 31. user (2026-06-22T09:49:06.754Z)

No exc\_info or traceback logging found

### 32. assistant (2026-06-22T09:49:08.607Z)

Let me also check if the run\_on\_vm.sh would expose this in any way:

### 33. user (2026-06-22T09:49:09.806Z)

```

39     fail_email() { # best-effort failure alert (the report email won't have fired on an
early crash)
40         echo "RUN FAILED ? attempting a failure alert"
41         if [ "$DO_EMAIL" = "true" ] && [ -d ~/rally ]; then
42             cd ~/rally 2>/dev/null && . .venv/bin/activate 2>/dev/null && \
43             NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
44             python - <<PY || true
45         from scripts.nightly_email import send_report
46         send_report("[KhelSutra nightly] ERROR ? engine regression VM run failed ($DATE)",
47             "The nightly engine-regression run failed on the VM before producing a
report.\n"
48             "See gs://$BUCKET/$REPORT_PREFIX/$DATE/ and the VM serial
log.\nGIT_SHA=$GIT_SHA", to="$EMAIL_TO")
49         PY
50         fi
51     }
52     trap 'fail_email' ERR
53
```

### 34. assistant (2026-06-22T09:49:12.194Z)

Perfect. Now I have a complete picture. Let me verify one more thing - whether the context manager behavior or other details could expose the password:

### 35. user (2026-06-22T09:49:26.330Z)

ERROR email failed (report still written): (535, b'5.7.8 Username and Password not accepted. For more information, go to\n5.7.8 <https://support.google.com/mail/?p=BadCredentials> d9443c01a7336-2c7439f8d1dsm76268095ad.51 - gsmtplib')  
 Simulating nightly\_reelcount.py call to send\_report:

```

=====
DEBUG: Exception would propagate: SMTPAuthenticationError
Caller caught: SMTPAuthenticationError

```

Password in logged output: False

```

=====

```

CONCLUSION:

1. The password variable in send\_report() is NOT in the exception message

2. The exception is caught by caller WITHOUT `exc_info=True`
3. The password is NOT exposed in logs CURRENTLY
4. HOWEVER: password is in the stack frame and COULD be exposed if:
  - Exception were logged with `exc_info=True`
  - Full traceback were printed
  - Debugger inspected the frame

### 36. assistant (2026-06-22T09:49:39.049Z)

Excellent. Now I have all the facts. Let me now compile my final assessment:

## VERDICT

The finding claims that the password "may leak in exception trace" when SMTP login fails. Let me verify the actual risk:

#### **\*\*WHAT THE CODE ACTUALLY DOES:\*\***

1. ``send_report()`` (lines 86-89 of `nightly_email.py`) has NO try-except around ``s.login(user, password)``
2. If login fails, ``SMTPAuthenticationError`` is raised and propagates up
3. The caller (`nightly_reelcount.py` lines 539-540) catches it with ``except Exception as e``
4. The caller logs with: ``log.error("email failed (report still written): %s", e)`` ? NO ``exc_info=True``

#### **\*\*PASSWORD EXPOSURE ANALYSIS:\*\***

- The password is NOT in the `SMTPAuthenticationError` message (only the SMTP server response is)
- The password variable is in the local scope of ``send_report()``
- Since logging does NOT include ``exc_info=True``, the traceback and local variables are NOT logged
- The password is currently NOT exposed in logs or console output

#### **\*\*THE LATENT ISSUE:\*\***

- If the exception were logged with ``exc_info=True`` OR if the traceback were printed, the local variable ``password`` would be exposed
- This is a hypothetical path, not the current code path
- The current implementation is defensive about this (by NOT using ``exc_info=True``)

#### **\*\*REAL PRODUCTION RISK:\*\***

- **\*\*Current state\*\***: Low. The password is not exposed in the current code path.
- **\*\*Potential risk\*\***: Medium. The lack of explicit error handling means uncontrolled exceptions could propagate and be logged with tracebacks elsewhere
- **\*\*Best practice\*\***: Wrap SMTP operations in try-except to explicitly prevent password variable exposure

**\*\*Is this a REAL defect?\*\*** Technically, the current code does NOT leak passwords in normal operation. However, it's a **\*\*latent security issue\*\*** because:

1. The password variable is in the stack frame and could be exposed if exception handling changes
2. It violates the principle of "secrets shouldn't be in local variables at the point of failure"
3. The best practice is explicit error handling to prevent any possibility of exposure

Let me call the StructuredOutput tool with my findings:

### 37. user (2026-06-22T09:49:48.583Z)

Structured output provided successfully

### 38. assistant (2026-06-22T09:50:02.908Z)

## Summary

I have **\*\*adversarially verified the SMTP security finding and REFUTED it as a real, production-**

impacting defect\*\*.

**\*\*Key findings:\*\***

1. **\*\*The claim is technically OVERSTATED\*\***: The finding states that "the password may leak in exception trace when logged or printed." This does not occur in current production code.

2. **\*\*Why the password is NOT exposed:\*\***

- SMTPAuthenticationError does NOT contain the password in its exception message (only SMTP server response)
- The caller at nightly\_reelcount.py:539-540 logs the exception WITHOUT `exc\_info=True`
- Without `exc\_info=True`, tracebacks (and their local variables) are not logged
- Verified via concrete testing: password does not appear in any logged output

3. **\*\*What IS a valid concern (defensive programming):\*\***

- The password variable exists in the stack frame of send\_report()
- If exception handling were changed to use `exc\_info=True` OR if tracebacks were printed, the password would be exposed
- Best practice would be to wrap SMTP operations in try-except to explicitly clear/sanitize error paths

4. **\*\*Severity: LOW (real=false)\*\***

- Current code does not leak secrets
- This is latent technical debt, not a production regression risk
- No immediate fix required, but explicit error handling in send\_report() would be more robust

**\*\*File paths with evidence:\*\***

- `C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/nightly-e2e/scripts/nightly\_email.py:86-89` ? SMTP login without try-except (but exception doesn't expose password)
- `C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/nightly-e2e/scripts/nightly\_reelcount.py:539-540` ? Caller catches exception without `exc\_info=True` (prevents traceback logging)